Lecture 5 - Bisimilarity

Matthew Farah, ⟨farahm11@mcmaster.ca⟩, 400468251

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- The *traces* of a process `P` to some state `p` in its LTS representation is the set of all sequences of actions which can be taken to reach `p`.
  - The traces of a process to its starting state is denoted by Trace(`P`).
- Two processes, `P`, `Q` are said to be *equivalent* if for some third process `R`, (`R` || `P`) = (`R` || `Q`).
  - `P` and `Q` are denoted as being equivalent by `P` ≡ `Q`.
- If `P` and `Q` are sequential or deterministic processes, then `P` ≡ `Q` $\iff$ Traces(`P`) = Traces(`Q`).
- However, in any other case, it is not sufficient for the traces of the two processes to be equal, since under parallel composition, two processes with the same traces can behave differently.
  - For example, for non-deterministic processes `P` and `Q` where Trace(`P`) = Trace(`Q`), it can be true that $(P \mid\mid R)$ can deadlock whereas $(Q \mid\mid R)$ does not for some process `R`.
- Hence, in the context of concurrent systems, equivalence is redefined as *bisimilarity*.
- Consider the LTS representations of `P` and `Q`:
  - Two states `p` ∈ `P` and `q` ∈ `Q` are said to be bisimilar if every action at `p` can be executed at `q`, and likewise if every action at `q` can be performed at `p`.
    - `p` and `q` being bisimilar is denoted `p` ≈ `q`.
    - Bisimilarity is both associative and commutative.
    - Important to note that a single state in `P` can be bisimilar to multiple states in `Q` and vice versa.
  - Two processes `P` and `Q` are said to be bisimilar if, for every state `p` ∈ `P` reachable by traces t, those some traces reach a state `q` ∈ `Q` which is bisimilar to `p`.
- `P` and `Q` are denoted as being bisimilar by `P` ≈ `Q`.
  - `P` ≈ `Q` $\implies$ (`P` || `R`) = (`P` || `R`) for all processes `R`.
- Bisimilarity cannot distinguish between two processes which can exhbit simultaneity or interleaving

Ex. Determine whether the following LTSs are bisimilar.

(a)

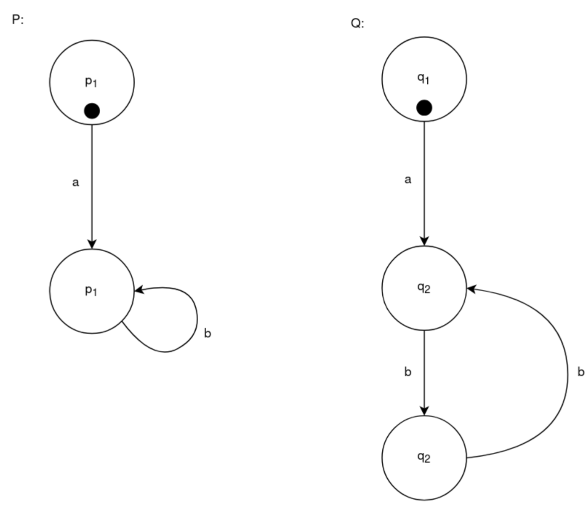

Figure 1: LTSs for part (a).

Solution:

They are bisimilar

(b)

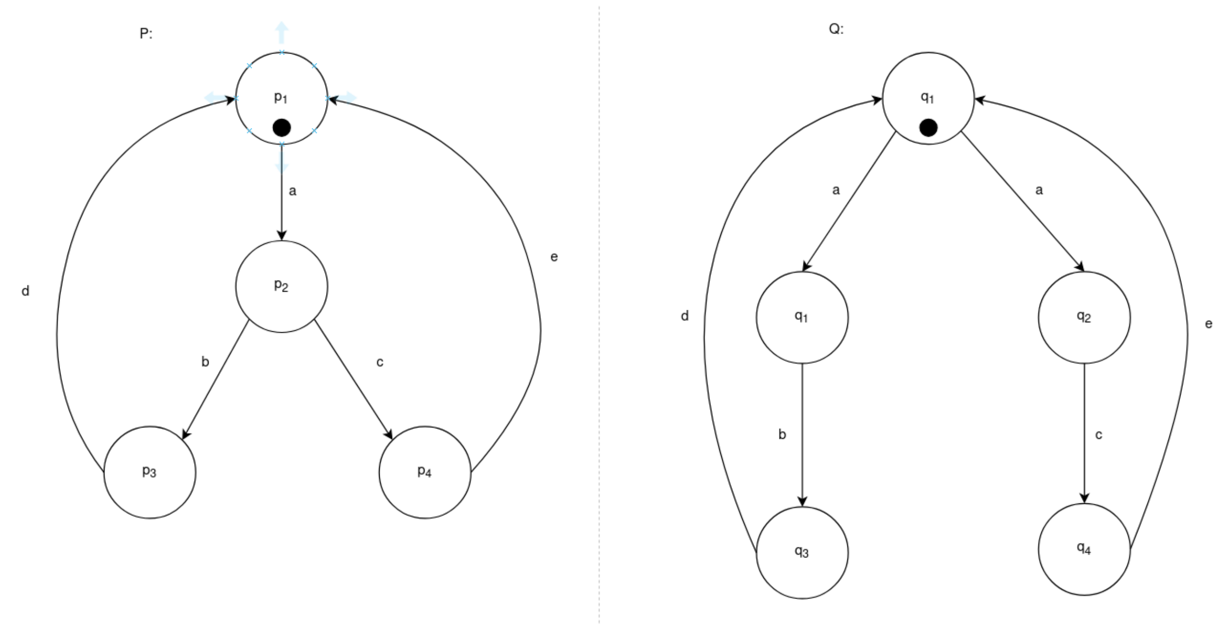

Figure 2: LTSs for part (b).

Solution:

They are not bisimilar